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Dual Simplex Method

- 1. Simplex Method
- 2. Big Method
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- 5. Graphical Method
- 6. F.S $\leftrightarrow$ B.F.S
convex sets
- 7. Revised Simplex method

Dual Simplex ^{Method} is a special variant of Simplex method where in each iteration optimality condition maintained.

$$\left. \begin{array}{l} \text{Min: } c_j - z_j \geq 0 \quad \forall j \\ \text{Max: } c_j - z_j \leq 0 \quad \forall j \end{array} \right\} \text{ optimality condition}$$

In this method, the initial solution may be infeasible. We need to remove the infeasibility by iterative table.

For Min Problem: If at iteration $c_j - z_j \geq 0$ is not maintained $\Rightarrow$ STOPPED

Example Min $Z = x_1 + 2x_2 + 3x_3$

Subject to

$$\begin{cases} 2x_1 - x_2 + x_3 \geq 4 \\ x_1 + x_2 + 2x_3 \leq 8 \\ x_2 - x_3 \geq 2 \end{cases}$$

$$\Rightarrow -(2x_1 - x_2 + x_3) \leq -4$$

$$\Rightarrow -2x_1 + x_2 - x_3 \leq -4$$

$$\Rightarrow -2x_1 + x_2 - x_3 + s_1 = -4 \quad (*)$$

$$\Rightarrow x_2 - x_3 \geq 2$$

$$\Rightarrow -(x_2 - x_3) \leq -2$$

$$\Rightarrow -x_2 + x_3 \leq -2$$

$$\Rightarrow -x_2 + x_3 + s_3 = -2 \quad (\otimes)$$

Min $Z = x_1 + 2x_2 + 3x_3 + 0 \cdot s_1 + 0 \cdot s_2 + 0 \cdot s_3$

Subject to

$$\begin{cases} -2x_1 + x_2 - x_3 + s_1 = -4 \\ x_1 + x_2 + 2x_3 + s_2 = 8 \\ -x_2 + x_3 + s_3 = -2 \end{cases}$$

$$x_1, x_2, x_3, s_1, s_2, s_3 \geq 0$$

and s_1, s_2, s_3 are slack variables

Initial Table

CB _i	C _j	1	2	3	0	0	0	Soln:
	B.V	x ₁	x ₂	x ₃	s ₁	s ₂	s ₃	
0	s ₁	-2	1	-1	1	0	0	-4
0	s ₂	1	1	2	0	1	0	8
0	s ₃	0	-1	1	0	0	1	-2
	Z _j	0	0	0	0	0	0	0
	C _j - Z _j	↑	2	3	0	0	0	

$C_j - Z_j \geq 0 \quad \forall j \rightarrow$ optimality condition is maintained.

$\Rightarrow$ optimal but not feasible.

$\Rightarrow$ Remove the infeasibility

Leaving variable:

check the most -ve value.

$\Rightarrow$ s₁ is the leaving variable.

Entering variable:

$$= \min \left\{ \frac{-(C_j - Z_j)}{y_{rj}} : y_{rj} < 0 \right\}$$

$$= \min \left\{ \frac{Z_j - C_j}{y_{rj}} : y_{rj} < 0 \right\}$$

$$= \min \left\{ \frac{-1}{-2}, -, \frac{-3}{-1}, -, -, - \right\}$$

$$= \min \left\{ \frac{1}{2}, 3 \right\}.$$

$$= \frac{1}{2}$$

$\Rightarrow x_1$ will enter to next basis.

Iteration - II

-2 is the key element

CB _i	C _j	1	2	3	0	0	0	Soln
	B.V	x_1	x_2	x_3	s_1	s_2	s_3	
1	x_1	1	$-\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	0	0	2
0	s_2	0	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{1}{2}$	1	0	6
0	s_3	0	-1	1	0	0	1	-2
	Z_j	1	$-\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	0	0	
	$C_j - Z_j$	0	$\frac{5}{2}$	$\frac{5}{2}$	$\frac{1}{2}$	0	0	

$$C_j - Z_j \geq 0 \quad \forall j$$

At this iteration, optimal but infeasible.

$\Rightarrow s_3$ will be the leaving variable.

Entering variable:

$$\min \left\{ \frac{Z_j - C_j}{y_{rj}} : y_{rj} < 0 \right\}$$

$$= \min \left\{ -\frac{5/2}{-1} \right\}$$

$$= \frac{5}{2}$$

$\Rightarrow x_2$ will enter to next basis.

$\Rightarrow -1$ is the key element.

Iteration - III

CB _j	C _j	1	2	3	0	0	0	Soln
	B.V	x_1	x_2	x_3	s_1	s_2	s_3	
		x_1	1	0	0	$-\frac{1}{2}$	0	$-\frac{1}{2}$
		s_2	0	0	3	$\frac{1}{2}$	1	$\frac{3}{2}$
		x_2	0	1	-1	0	0	-1
		z_j	0	2	-2	$-\frac{1}{2}$	0	$-\frac{5}{2}$
		$C_j - Z_j$	1	0	5	$\frac{1}{2}$	0	$\frac{5}{2}$

$$C_j - Z_j \geq 0 \quad \forall j$$

$\Rightarrow$ optimal condition is maintained
and the soln is feasible

$$x_1 = 3, \quad x_2 = 2, \quad x_3 = 0$$

$$s_2 = 3, \quad s_1 = 0$$

$$\boxed{\text{Min } Z} = x_1 + 2x_2 + 3x_3 + 0 \cdot s_1 + 0 \cdot s_2 + 0 \cdot s_3$$

$$= 3 + 2 \cdot 2 + 3 \cdot 0 + 0$$

$$= 7.$$